

# ModelSystemModding\_Documentation

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**Document Version:** 1.1

## Change History

| Version | Changes                                                      |
|---------|--------------------------------------------------------------|
| 1.1     | Added section with additional details about the Editor tool. |

## 1 Introduction

The purpose of this document is to define some of the information modders need to override or add to the model system. View the other accompanying documents for specific information about relevant file formats.

Please note that there are limitations on how model modding can work.

- Changing model mods that are loaded requires a game restart to take effect.
- There may be unstable situations due to the broad range of conditions that developers may not have used.

General advice:

- Keep models simple whenever possible to keep them cheap for the system to use.
- Destruction models in particular can be costly if they have high poly/chunk counts.

## 2 Converting Models

You can convert models from FBX formats into TMM and TMA files that the game uses in the editor. This conversion is performed by defining fbximport and destruction import data as required by the type of model you are trying to convert.

- All **\*.fbximport** and **\*.destructionimport** files have been bundled into **ArtImportFilesForModders.bar** located under **game/Art** in your install directory as examples. Note that this does not include the **FBX** files, so they are mostly just examples for those curious about how a specific thing was achieved for some model.
- A sample of some more complete examples that include the associated **FBX** files can be found in **ArtExamplesForModders.bar**, which is also located under **game/Art**.

To convert a model, follow these steps:

1. Create a local mod.
2. Add the FBX file/s you want to convert to the local mod folder.
3. Create the metadata files matching \*.fbximport (and \*.destructionimport if needed).
4. Launch the game.
5. Open the Editor.

6. Navigate to Objects > Model Convert Tool.
7. Find the model you want to convert in the list and confirm to convert.
8. The output (if successful) will be the resulting \*.TMM and \*.TMM.data files if it was a model or the \*.TMA file if it was an animation.

Note: this will also work with Speed Tree XMLs if set up in suitable ways.

### 3 Adding New Models to Data

There are a couple of places that will require additions to bring a model in-game. These steps assume you have a local mod where you have created models that you want to add.

1. Ensure you have converted the model as defined in section 2.
2. Ensure you have a \*.material file matching the \*.TMM file name if you are making a model.
3. Ensure you have any AnimXMLs and other associated files created.
4. Create a **modelManifest\_mods.xml** file in “**intermediate/modelcache/**”
  - a. Parent node should be a <modelCache> node.
  - b. Child nodes should be <file> nodes with the relative path to the TMM file, including the .tmm extension.
  - c. This tells the game during startup that the model exists.

Example:

```
1 <modelCache>
2   <file
   mergeMode="add">norse\units\siege\ballista_test\ballista_test.tmm</file
3 >
4 </modelCache>
```

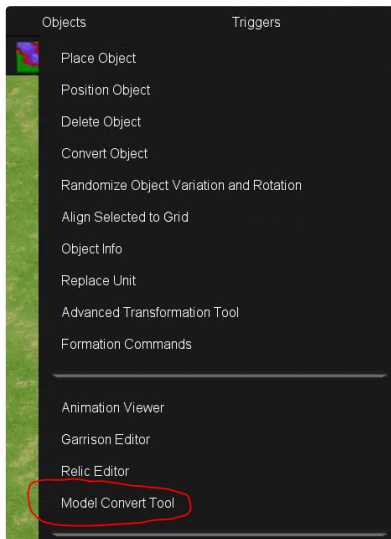
5. If you are creating a new unit, then you may want to add a proto\_mods.xml file under game/data/gameplay.
6. Create a **simdata\_mods.xml** file in “**game/data/gameplay/**”
  - a. Parent node should be a <simdatamods> node.
  - b. You can define <visual> nodes that use merge modes on the file and type. Review the SimDatabase documentation for more details about what these should contain.
  - c. You can define <animxml> nodes that use merge modes in the file. Review the SimDatabase documentation for more details about what these should contain.

Note: Currently, this does require a lot of hand editing to create the mods. It is recommended that a tool be used to generate this data to reduce overhead. The developer version of this tool is not ready yet for modder use, but it may become available in the future.

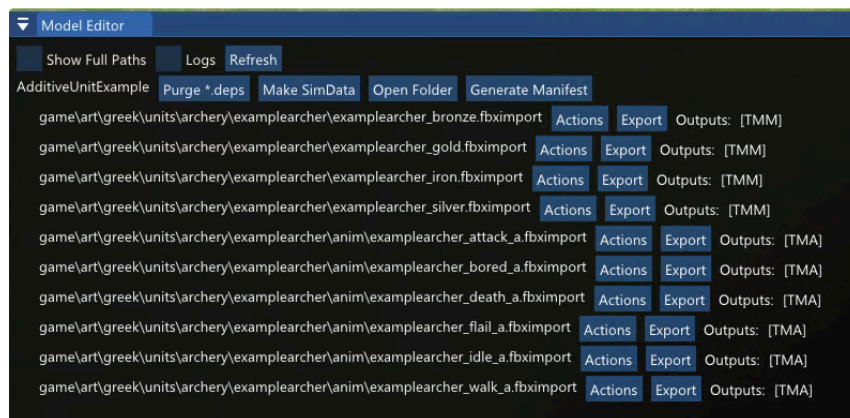
7. Launch the game and test if your model works.

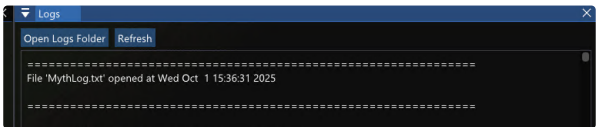
### 4 Editor Tool Overview

Open the Tool from the Objects > Model Convert Tool menu option.



You will see a UI that looks similar to that below if you have one or more local mods. Each mod will appear listed in alphabetical order with all fbximport files listed below.



| Option / Button | Description                                                                                                                                                                                                                                                                     |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Show Full Paths | When enabled this will show full paths in the UI for the mod folder path and for the fbximport and output path.                                                                                                                                                                 |
| Logs            | <p>Ticking this will show another separate UI that contains the content of mythlog.txt with a button to open the folder containing the logs and a button to refresh the content shown.</p>  |
| Purge *.deps    | When you convert to the TMM/TMA formats it will also output a .deps file that defines the dependency data with a hash of each file. This is a full path to the file location so you may wish to delete these files before submitting a mod.                                     |

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Make SimData                              | <p>This button will generate a SimData_Generated.xml file in the local mod directory. It is recommended you delete this file before submitting a mod. The purpose of this is to allow you to copy parts out of the file to make creation of the SimData_Mods.xml file easier.</p> <p>Generating this data does output some additional debug data that can be viewed in the Logs view.</p>                                 |
| Open Folder                               | Opens the folder for the local mod.                                                                                                                                                                                                                                                                                                                                                                                       |
| Generate Manifest                         | <p>Generates the modelmanifest_mods.xml file for you based on all TMM and TMA files that exist in your local mod. This automatically excludes listing files that are part of the base game.</p> <p>Note that this will generate an file with no child nodes if you have no TMM/TMA files in your mod so it is only recommended to generate this if you need to populate the file with additive model file references.</p> |
| Actions > Copy Relative Path to Clipboard | <p>Copies a version of the path that is useful for pasting into places you may use the model in other files such as the AnimXML files.</p> <p>Example:<br/>game\art\greek\units\archery\examplearcher\example archer_bronze</p>                                                                                                                                                                                           |
| Actions > Copy Full Path to Clipboard     | <p>Similar to above except it is an absolute full path to the fbximport. Useful if you need to edit the import file.</p> <p>Example: C:\Users\&lt;YOURUSER&gt;\Games\Age of Mythology Retold\<br/>&lt;YOURSTEAMID&gt;\mods\local\AdditiveUnitExample\game\art\greek\units\archery\examplearcher\example archer_gold.fbximport</p>                                                                                         |
| Actions > Copy Export Path to Clipboard   | <p>Similar to above except it is an full path to the export location. Useful for knowing where the file will end up.</p> <p>Example: C:\Users\&lt;YOURUSER&gt;\Games\Age of Mythology Retold\<br/>&lt;YOURSTEAMID&gt;\mods\local\AdditiveUnitExample\intermediate\modelcache\greek\units\archery\example archer\examplearcher_bronze</p>                                                                                  |
| Export                                    | <p>Attempts to convert from the FBX Import and associated dependent files into a valid TMM/TMA file. The result will be visible in a Status message or in the Logs output.</p>                                                                                                                                                                                                                                            |

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Outputs | <p>This lists the current state of the outputs. It will say if there are none and if the result files include any of the following:</p> <ul style="list-style-type: none"><li>• [TMM] = A TMM file exists at the export location with the relative path name.</li><li>• [TMA] = A TMA file exists at the export location with the relative path name.</li><li>• [HKT] = A HKT file exists at the export location with the relative path name. (This is a physics data file).</li><li>• [_death.tmm] = An extra associated death version file.</li></ul> <p>If there is not at least one of TMM or TMA for the files it will show:</p> <ul style="list-style-type: none"><li>• [No TMM/TMA]</li></ul> |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|